

Criterion E: Evaluation

Evaluation of the Product

1. Bill can deposit and withdraw money from his virtual account balance without any disruptions with the calculations or input.

Mostly met. While the functions generally allow for withdrawals and deposits without errors or disruptions, Bill noticed that withdrawal of inputted negative values can happen.¹

2. Bill can access the history of his account and each data. Basically Bill can log in his account and access his previously created budget and account balance despite shutting down the application.

Met. Bill can log-in his account without any issues and see his previously created budget.

3. Bill can create new or delete existing budgetary data regarding categories, subcategories, dates of each category/subcategory to be paid, and the amount to be paid

Met. Add and remove category button act as intended, though an update button for updating the values of budgetary data of each row would allow better usability.

4. Bill's password within the database is to be obfuscated

Met. Bill's password is hashed and salted, though he would like to see his original password later as it is not commonly used by him.

5. Bill has the choice to send his created budget to his gmail account

Met. Functions act as intended, though he would also like to send graphs or date to run out of money to his email.

6. The sqlite database is to be constantly saved to a Google Drive folder using a Python API

¹ See Appendix A

Met. Everytime Bill closed the program, a new updated database file appears in the database

7. Bill is able to sort budgetary data (e.g categories, subcategories, amounts to be paid, dates to be paid) either in ascending or descending order based on his preference and importance of each

Met. Based on column header clicked, rows are sorted either normally or in reverse

8. Password strength policies (e.g. upper and lower case letters, special characters) for password creation are to be taken into consideration.

Met. Though these policies functioned correctly, error messages were sometimes confusing depending on user input validation.

9. Projections of when Bill's money will run out based on his average spending, payment dates for each category/subcategory, and the corresponding amounts should be provided.

Met. Though it would also be very beneficial if spending targets could be set based on average spending.

10. Graphs showcasing his spending trends against dates to be paid, amounts, and his average spending should also be provided and plotted based on Bill's command

Met. Though other types of diagrams are more preferable

11. Creation of an SQLite database where Bill's budgetary data and account information are to be updated and stored based on changes made within the application (e.g Bill adds a new category)

Met. A database is initialized and operable.

12. Any further extreme data, abnormal data, and exceptions are to be handled

Mostly met. Due to the issue with the withdraw function.

Recommendations for Further Development

1. Creation of spending targets: This functionality would allow Bill to determine how much he intends to spend within a specific period or category, helping him manage his budget more effectively. Alerts could also be provided when spending approaches or exceed the set targets as he attempts to create a new row of budgetary data, ensuring better financial control.
2. Sending graphs and financial projections to email address: To allow for further organization and storing of data within gmail, Bill suggested implementing the send email function for the financial projection tab. To help facilitate this, new buttons should be created which utilize the EmailMessage API to send images and other types of data other than text to his email².
3. Utilization of different diagrams based on preference: To mitigate the issue of confusing line diagrams³, different types of diagrams can be chosen by Bill based on his preference, such as scatter diagrams. These can be created using the matplotlib library.

Word Count: 354

Overall Word Count: 1987

² Vfc, Jerry. "Sending Images through Email." *Stack Overflow*, 18 Aug. 2022, stackoverflow.com/questions/73397859/sending-images-through-email.

³ See Appendix A and Criterion D

Bibliography

Vfc, Jerry. "Sending Images through Email." *Stack Overflow*, 18 Aug. 2022,
stackoverflow.com/questions/73397859/sending-images-through-email.